PROJECT mC

VIBEGUARD SEMESTER IMPLEMENTATION AND FUTURE RESEARCH ROADMAP

Frozen Level-1 Build Plan, Validation Programme and Post-Semester PIRG Research Path

Course	PBCST504 - Microcontrollers Micro Project
Programme	B.Tech Computer Science and Engineering (Cyber Security)
Group	Group 8
Project Coordinator	Ms. Sagna L T, Assistant Professor, CSE (CY)
Prepared for	Semester implementation approval and execution
Date	03 August 2026
Document status	Teacher-facing implementation dossier; controlled draft

Controlling decision

This dossier implements only the frozen VibeGuard Level-1 semester architecture: one rigidly mounted ADXL345-class accelerometer, one ESP32, local normal-versus-induced-abnormal classification, RGB indication and USB evidence. PIRG remains a separate, unbuilt post-semester research hypothesis.

Patent boundary

The ordinary semester build is conventional and is not represented as a patent-ready invention. Any future filing decision requires new physical evidence, refreshed prior-art work, institutional IPR review and professional advice.

Prepared from the controlled Project mC authority set and the Master Evidence, Costing and Decision Basis. This document is an engineering implementation plan, not legal advice, a patent opinion or freedom-to-operate clearance.

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Document control and approval use

Control item	Value
Architecture baseline	Frozen VibeGuard Level-1, single-node ADXL345-class sensor + ESP32
Expected semester budget	Approximately ₹2,200
Protected conservative envelope	Approximately ₹3,900; below the ₹5,000 ceiling
Core implementation window	Reach minimum demonstrable system by Week 8; Weeks 9-12 reserved for validation, recovery and presentation
Future research	PIRG only after semester foundation; separate folder, budget, evidence and confidentiality
Primary project record	This dossier becomes the controlling build plan after teacher approval

Approval checkpoints required before procurement freeze

1. Teacher approval of the change from OpenBraille to VibeGuard and the exact Level-1 semester scope.
2. Confirmation whether the effective build window is approximately eight weeks or the full twelve-week plan.
3. Approval of the selected ESP32 board variant and procurement source.
4. Faculty/laboratory safety review of the powered motor, shaft guard and retained eccentric-mass fixture.
5. Approval of a quantitative Level-1 held-out validation protocol before the locked test.

Executive implementation decision

Recommended semester build

Build VibeGuard as a low-cost, single-node embedded vibration-monitoring prototype that distinguishes a controlled normal rig state from a repeatable induced-abnormal state. Use RMS plus persistence as the minimum classifier; add triggered FFT/bands and ordinary Mahalanobis only as comparisons after the baseline works.

VibeGuard is selected because its main risks - sensor authenticity, rigid coupling, stable acquisition, repeatable fault injection, threshold selection and data leakage - can be isolated through scripted experiments and logs. Its expected budget leaves more recovery headroom than OpenBraille, and the team's strongest capabilities align with ESP32 firmware, signal processing, USB evidence and quantitative validation.

The project should not be described as a complete industrial predictive-maintenance product. The correct semester claim is a controlled proof that a low-cost embedded node can acquire vibration, establish a normal condition, detect a deliberately introduced abnormal condition and report the result locally without cloud processing.

Success definition at a glance

Layer	Required semester result	Evidence
Hardware	Stable ESP32 + ADXL345 SPI acquisition on a rigid target mount	Acceptance checklist, axis verification, ODR record, dropped-block and saturation counters
Rig	Safe, repeatable normal and induced-abnormal operating states	Rig configuration IDs, photos, RPM/load notes, retained eccentric mass and guard inspection
Minimum decision	RMS threshold with persistence works locally	Frozen calibration, state logs, classification latency and held-out confusion matrix
Secondary analysis	FFT/bands and ordinary Mahalanobis are compared without delaying the baseline	Feature plots/tables, resource use and comparator results

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Layer	Required semester result	Evidence
Output	Clear RGB state and USB evidence	Live demonstration and exportable CSV/log file
Honesty	Limitations and failed runs are preserved	Negative-results log, remount records and scope statement

1. Purpose, audience and scope

1.1 Purpose

This document converts the approved Phase 3C VibeGuard architecture into a procurement-ready and execution-ready semester plan. It defines the exact hardware, interfaces, mechanical test rig, firmware modules, validation controls, team responsibilities, budget, schedule, risks, outputs and future-research boundary.

1.2 Audience

- Project coordinator and teacher: approval, scope control, safety and review gates.
- Five-member student team: subsystem ownership, integration sequence and evidence requirements.
- Department laboratories: equipment booking, safety assistance and optional measurement support.
- Institutional innovation/IPR reviewers: separation of the public semester prototype from confidential future research.

1.3 Scope hierarchy

Scope tier	Included	Not included
Semester Level 1	One target-mounted sensor; normal versus induced abnormal; local processing; RGB and USB evidence	Mixed-source attribution, field deployment, cloud platform, RUL or patent claim
Post-semester Level 2	PIRG target/interferer research after a stable Level-1 platform exists	Assumed success or automatic filing
Commercial/industrial	Only after later validation and professional review	Any promise in the current project

2. Engineering identity and problem definition

2.1 Engineering identity

VibeGuard is a passive edge condition-monitoring system. Its defining feature is not the accelerometer itself, but the local interpretation of vibration on constrained embedded hardware. It targets educational and small-enterprise contexts where expensive industrial monitoring infrastructure may be unavailable.

2.2 Semester engineering hypothesis

> **Hypothesis to test**
>
> A low-cost target-mounted accelerometer and ESP32 can consistently distinguish a controlled normal operating condition from a deliberately introduced abnormal-vibration condition using lightweight local analysis, without cloud processing.

2.3 What the hypothesis does not claim

- It does not identify a real bearing defect or predict remaining useful life.
- It does not prove that one sensor can attribute vibration to the correct machine in a dense factory.
- It does not establish universal thresholds across motors, mounts or environments.
- It does not claim that FFT, Mahalanobis distance or edge execution is novel.

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3. Frozen system architecture

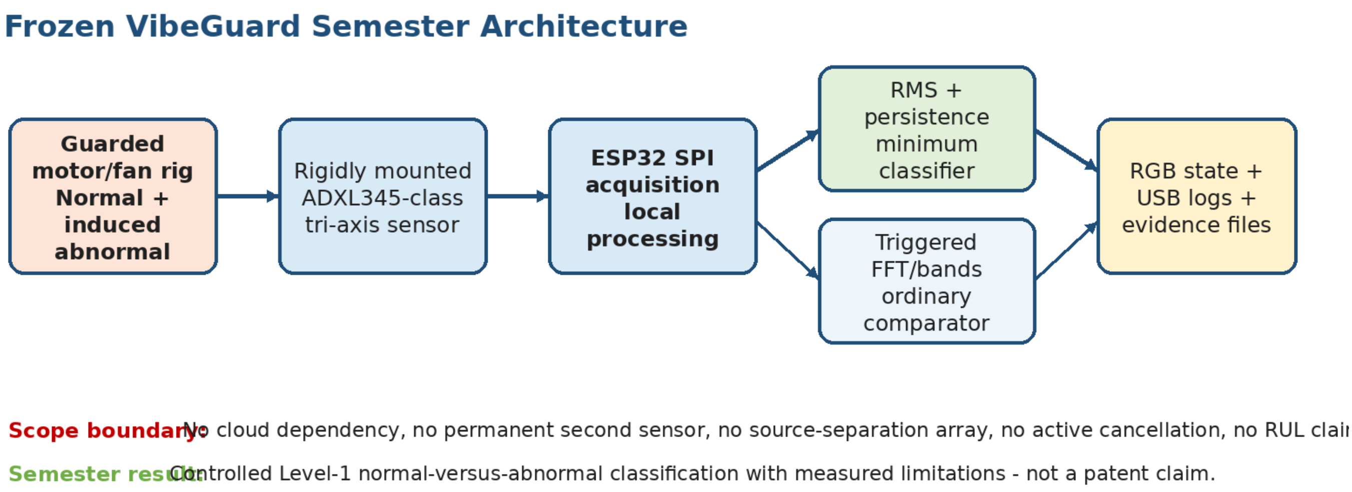

Figure 1. Frozen Level-1 architecture and explicit semester scope boundary.

3.1 Subsystem boundaries

Subsystem	Inputs	Outputs	Owner acceptance
Mechanical target rig	12 V power, normal/abnormal fixture state	Repeatable vibration transmitted to target housing/base	Guarded, stable, labelled and reproducible
Sensor/mount	Mechanical vibration, 3.3 V power, SPI clock/control	Timestamped X/Y/Z samples	Correct axes, no saturation, documented mount ID
Embedded acquisition	SPI samples and interrupt/FIFO state	Raw windows, integrity counters	Stable ODR, zero block loss in locked run
Feature pipeline	Raw windows	Time features, triggered spectra/bands	Versioned, deterministic and reproducible
Decision/state	Calibrated baseline and current features	Calibrating/Normal/Abnormal state	Frozen thresholds and persistence
Human/evidence interface	State and records	RGB indication, USB CSV/logs	Teacher-readable demonstration and audit trail

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4. Detailed hardware architecture and BOM

4.1 Required purchased and fabricated items

Item	Qty	Exact semester role	Expected	Conservative	Acceptance/ procurement rule
ESP32 development board	1	Main controller: SPI acquisition, local processing, RGB and USB	₹400	₹899	Verified WROOM-32 DevKit baseline; ESP32-S3 is authorized fallback. Confirm pinout, USB-UART and regulator.
ADXL345 breakout	1	Rigidly mounted tri-axis vibration sensor	₹229	₹350	Buy from a reputable source; verify board layout, axes, ODR, noise and saturation before integration.
12 V geared motor or suitable fan	1	Controlled target test article	₹215	₹300	Choose a stable mounting face and guarded shaft. RPM is metadata, not the differentiator.
12 V, 1 A regulated adapter	1	Separate motor-rig supply	₹140	₹330	Confirm polarity, connector and current margin.
Common-cathode RGB LED + resistors	1 set	Green Normal, Blue Calibrating, Red Abnormal	₹25	₹65	Three separate LEDs are acceptable if clearer.
Perfboard	1	Stable final sensor/controller interface	₹58	₹85	Breadboard only for bring-up; final SPI wiring should be short and strain-relieved.
Wires, headers and connectors	1 lot	SPI, low-voltage power and USB-support connections	₹180	₹250	Separate motor-current paths from sensor/logic paths.
Heavy base and rigid sensor mount	1 set	Repeatable mechanical coupling and safe support	₹300	₹500	Local allowance; preserve CAD/measurements, photos and mount ID.
Eccentric mass, guard and fasteners	1 set	Safe, repeatable induced abnormal condition	₹200	₹350	Mass must be mechanically retained and operated behind a guard.
Passives and protection	1 lot	Decoupling, switch, DC jack, fuse/protection	₹150	₹280	Finalize ratings after motor procurement.
Delivery/replacement allowance	1	Logistics and one controlled rework margin	₹300	₹500	Not a hidden feature budget.

Budget decision

Expected semester planning total: approximately ₹2,200. Protected conservative envelope: approximately ₹3,900. The future PIRG rig, second motor, temporary reference instruments, patent search and professional fees are not included.

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4.2 Reusable or institutional equipment

Equipment	Use	Required status
Laptop with Arduino IDE and Python	Firmware, serial logs, data analysis and reports	Assumed available
Soldering station and hand tools	Final perfboard and connectors	Book before Week 3
Digital multimeter	Supply, continuity and basic current checks	Required
Oscilloscope or logic analyzer	SPI timing, rail noise and dropped-sample debugging	Strongly useful; borrowed
Bench supply with current limit	Safe motor and controller bring-up	Strongly recommended
Tachometer/RPM reference	Rig metadata and repeatability	Useful but optional
Calipers and fabrication tools	Bracket, guard and base measurements	Required for mount repeatability
Camera	Mount/remount evidence and demo recording	Required evidence tool

5. Mechanical test rig and safety

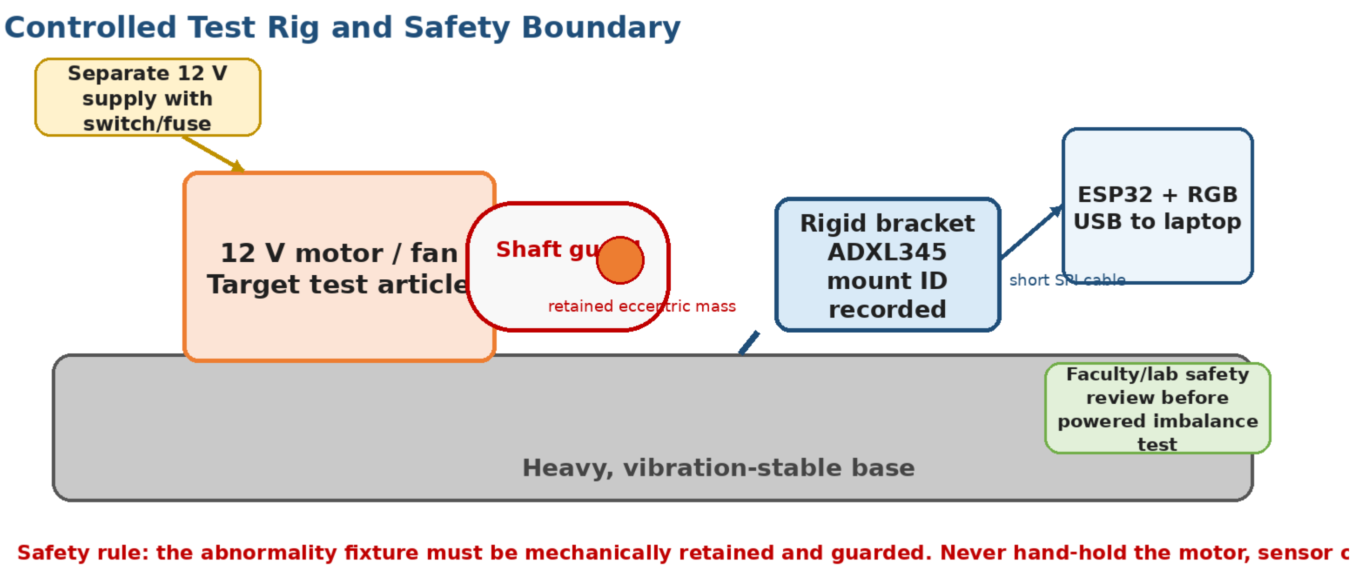

Figure 2. Proposed controlled rig. The final dimensions and motor variant remain procurement-dependent.

5.1 Rig design rules

1. Mount the motor/fan to a heavy base using bolts or a rigid bracket. Do not rely on hand pressure, foam tape or a loose table surface.
2. Mount the ADXL345 rigidly to the target housing or a documented target bracket. Magnetic, foam or soft adhesive mounts are prohibited for the final evidence run.
3. Create the abnormal state with a small, retained eccentric mass or another repeatable safe method. Record mass, radius, position, speed/load and configuration ID.
4. Install a physical shaft guard and an emergency power disconnect before abnormal operation.
5. Run first at low voltage/speed or under current-limited supply. Inspect fasteners and the eccentric fixture before every session.
6. Keep the ESP32 and sensor wiring outside the rotating hazard area and strain-relieve the short SPI connection.

5.2 Safety stop conditions

- Visible movement of the motor base or guard.
- Loosening, cracking or movement of the eccentric mass or shaft fixture.

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- Abnormal heating, smell, sparking, adapter instability or repeated resets.
- Sensor/bracket detachment or wiring entering the rotating zone.
- A faculty/lab reviewer requests suspension or redesign.

6. Electrical integration and provisional pin map

6.1 Power domains

Domain	Source	Loads	Rule
Logic/sensor	ESP32 USB input and 3.3 V regulator	ESP32, ADXL345, RGB LED	Low-voltage only; local decoupling at sensor; do not power the motor from USB or ESP32 regulator
Motor rig	Separate 12 V adapter with switch/protection	Motor/fan only	Keep motor-current wiring separate. A common ground is not required unless a later controlled interface explicitly needs it.
Laptop/USB	Laptop USB	ESP32 programming and logs	Data/control only; do not use as the motor power path

6.2 Provisional ESP32-WROOM-32 DevKit pin assignment

> **Implementation note**
>
> This is a Phase 4 wiring proposal, not a source-file mandate. Revalidate every pin against the exact purchased board. ESP32-S3 uses a different approved pin map.

Signal	ADXL345 / output	Provisional ESP32 pin	Notes
3.3 V	VCC	3V3	Use local 0.1 µF decoupling at the breakout
Ground	GND	GND	Short return path; avoid motor-current routing
SPI clock	SCL/SCLK	GPIO 18	Start at a conservative clock; increase only after integrity testing
SPI MOSI	SDA/SDI	GPIO 23	ESP32 to sensor
SPI MISO	SDO	GPIO 19	Sensor to ESP32
SPI chip select	CS	GPIO 5	Dedicated output
Data-ready interrupt	INT1	GPIO 4	Optional but recommended for stable timing
RGB green	LED + resistor	GPIO 25	Common-cathode LED or separate indicator
RGB blue	LED + resistor	GPIO 26	Calibrating
RGB red	LED + resistor	GPIO 27	Abnormal/fault indication
USB serial	On-board USB-UART	Laptop USB	Logging, commands and firmware upload

6.3 Wiring acceptance checklist

- Board powers without motor connected; 3.3 V rail measured and stable.
- ADXL345 device ID and axis response verified before high-rate acquisition.
- SPI wiring kept short; final evidence build soldered or locked in a rigid connector.

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• No sensor saturation during normal or induced-abnormal pilot runs.
• USB logs remain stable while the motor is switched and while the abnormal state is active.

7. Firmware and software architecture

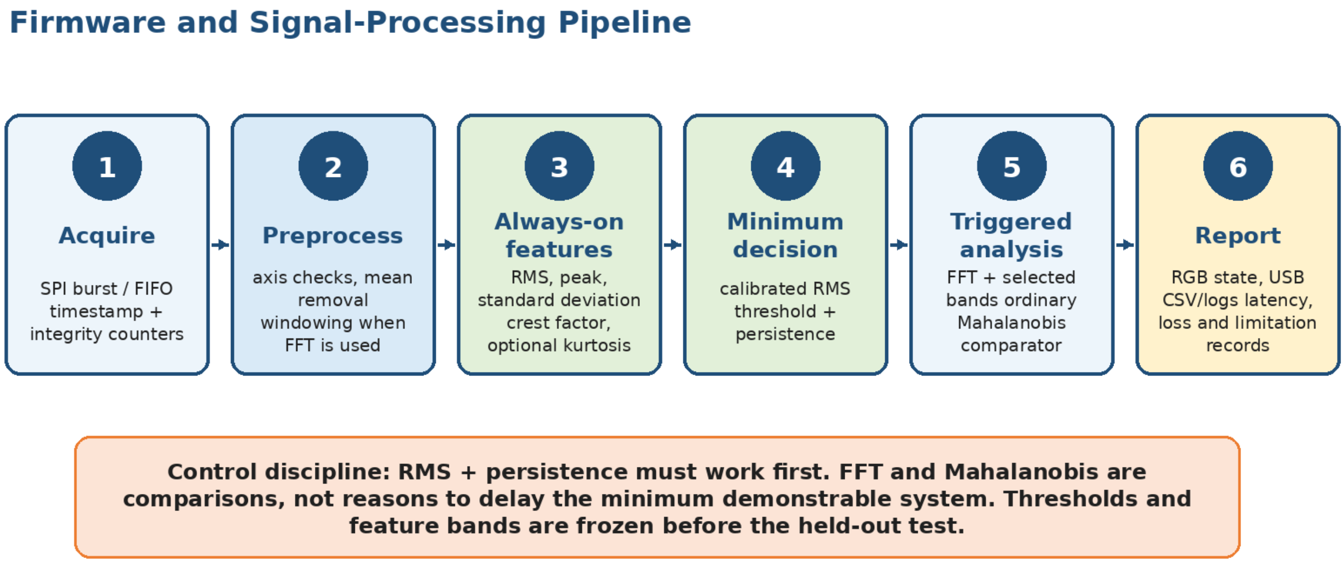

Figure 3. Staged pipeline. The minimum RMS/persistence path controls schedule risk.

7.1 Firmware module breakdown

Module	Responsibility	Required outputs/tests
Configuration manager	Board variant, pin map, ODR, range, window size, threshold version and build ID	Human-readable configuration block printed at startup and stored with logs
ADXL345 driver	Initialization, device ID, range/ODR, burst/FIFO reads and interrupt status	Axis test, known orientation check and read-error counter
Acquisition engine	Timestamp samples/windows and maintain loss/overrun counters	Measured achieved sample rate; zero locked-run block loss
Preprocessor	Mean/DC removal; optional Hann window for FFT; scaling and validation	Deterministic output verified against offline script
Time-feature engine	RMS, peak, standard deviation, crest factor and optional kurtosis	Unit tests on stored windows
Persistence classifier	Normal baseline, threshold and consecutive-window logic	Calibrating/Normal/Abnormal transitions and threshold log
Triggered FFT/bands	Compute spectrum only when scheduled or triggered	Frequency-resolution and latency record
Mahalanobis comparator	Regularized normal-space score; optional comparison only	Offline/embedded agreement and RAM/timing record
State/output manager	RGB state and fault behavior	Visible deterministic indication
USB logger/command interface	CSV logs, metadata, start/stop/calibrate commands	Recoverable files with schema/version
Health monitor	Saturation, read errors, timing overruns and configuration faults	No silent acquisition failure

7.2 Repository and configuration structure

Folder/file	Purpose
firmware/	ESP32 source, libraries, board configuration and release tags

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Folder/file	Purpose
hardware/	Wiring record, BOM, datasheets, board photos and mount CAD/dimensions
rig/	Motor, base, guard and abnormality configuration IDs
data/raw/	Immutable raw or minimally transformed session files
data/processed/	Versioned feature exports generated by scripts
analysis/	Python notebooks/scripts, metrics and plots
tests/	Unit tests, acceptance checklists and locked-run protocol
docs/	Teacher reports, weekly reviews, risk and decision logs
future_pirg_private/	Access-restricted future-research material; not used in semester grading

8. Signal-processing and decision logic

8.1 Acquisition parameters to resolve in pilot testing

Parameter	Starting proposal	Decision rule
Range	Start at ±4 g full-resolution mode	Increase only if saturation is observed; record every change
Output data rate	Start at 800 Hz; evaluate 1,600 Hz if spectral evidence requires it	Use the lowest rate that captures discriminating energy with stable SPI timing
Window	256 samples at 800 Hz or equivalent duration	Freeze before locked validation; overlap must be reported
FFT	256 or 512 points with Hann window	Triggered/periodic comparison; not always-on if it threatens acquisition
Normal calibration	Multiple complete normal sessions, not one short run	Separate calibration/training sessions from held-out sessions
Threshold	Derived from normal distribution with owner-approved rule	Do not change after viewing held-out results
Persistence	Require abnormal evidence for consecutive windows/time	Select on calibration data; report classification latency
Bands/features	Select from pilot evidence and physical plausibility	Freeze the list before held-out testing

8.2 Minimum classifier

The minimum classifier is a calibrated RMS threshold with persistence. The firmware calculates the vibration magnitude or a predeclared axis combination, estimates a normal baseline from approved calibration sessions, and raises the Abnormal state only when the threshold is exceeded for the approved persistence period. A single transient shock should not automatically become a persistent machine fault.

8.3 Triggered FFT and band features

FFT analysis is used to explain and compare the state change, identify repeatable spectral components and support later research. It is triggered when the low-cost stage crosses a pre-trigger or at scheduled diagnostic intervals. The report must disclose sample rate, window length, frequency resolution, overlap and selected bands.

8.4 Ordinary Mahalanobis comparator

Mahalanobis distance may be implemented as an optional normal-space comparator over a small feature vector. It must use regularized covariance and must be benchmarked against the simpler RMS/persistence baseline. The semester project succeeds without it if the minimum classifier and evidence programme are complete.

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8.5 State-machine behavior

State	Entry condition	Output	Exit/exception
STARTUP/SELF-CHECK	Power-on, configuration and sensor identity checks	Blue pulse / USB status	Fault if sensor or configuration is invalid
CALIBRATING	Approved normal session and stable acquisition	Blue	Restart if saturation, movement or timing fault occurs
NORMAL	Features inside calibrated bounds	Green	Abnormal evidence enters persistence counter
ABNORMAL	Threshold/persistence rule satisfied	Red	Return only after approved recovery/hysteresis rule
FAULT/INVALID	Sensor read error, saturation, timing loss or mount-invalid condition	Logged fault; distinct visible pattern if implemented	Requires operator action or controlled restart

9. Data and evidence management

9.1 Required session metadata

- Project, firmware and configuration version.
- Sensor board identity, axis orientation, range, ODR and SPI clock.
- Mount ID, photographs, fasteners/adhesive method and whether the sensor was remounted.
- Motor/fan identity, supply voltage, speed/load proxy and abnormality-fixture configuration.
- Session date/time, operator, target state and any transient events.
- Sample count, lost blocks, read errors, saturation count and timing statistics.
- Threshold/feature version and whether the session was calibration, development or locked hold-out.

9.2 Data split rule

Leakage prevention

Never randomly split windows from the same continuous run into training and test sets. Hold out complete sessions, days and remounts so the result tests repeatability rather than memorizing one recording.

9.3 Evidence package for final review

Evidence file	Minimum content
Configuration record	Exact board, sensor, pin map, sample rate, range, window, feature and threshold version
BOM and procurement record	Seller, date, unit cost, delivery and acceptance result
Rig record	Annotated photos, dimensions, safety inspection and state definitions
Session register	Every run, including failures and exclusions with reasons
Raw/processed data	Immutable original files and reproducible processing scripts
Validation report	Confusion matrix, false alerts, missed detections, latency, resource use and limitations
Demo checklist	Power-up, calibration, normal run, abnormal run, RGB/USB evidence and shutdown
Contribution record	Dated member responsibilities, code commits, hardware changes and experiment ownership

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10. Validation and acceptance programme

10.1 Validation stages

Stage	Question	Minimum evidence	Stop/fallback
V0 - Component acceptance	Are the purchased ESP32, sensor, motor and supply usable?	Device ID, axis response, stable rails, motor/guard inspection	Replace or redesign the failing component before integration
V1 - Acquisition integrity	Can the system collect stable timestamped data?	Achieved ODR, error/loss counters, no unexplained saturation	Reduce SPI clock/ODR, shorten wiring or move to perfboard
V2 - Normal repeatability	Is the controlled normal state stable across sessions?	Multiple sessions and remount record; feature distributions	Fix rig/mount before classifier tuning
V3 - Abnormal repeatability	Is the induced state safe and distinct?	Repeated labelled abnormal sessions and physical metadata	Redesign the fixture; do not tune around an unstable fault
V4 - Minimum classifier	Does RMS + persistence work locally?	Frozen threshold, latency and development results	Simplify features or improve rig labels
V5 - Comparator evaluation	Do FFT/bands or Mahalanobis add useful evidence?	Same held-out protocol; resource comparison	Drop optional method if it adds no value
V6 - Locked held-out run	Does the frozen system generalize to untouched sessions/remount?	Pre-registered metrics; no threshold changes	Report failure honestly; retain project as measurement study
V7 - Demo and documentation	Can another person operate and explain the project?	Checklist, rehearsal and source-backed report	Freeze features and repair only

10.2 Proposed Level-1 acceptance targets for teacher approval

Status of numbers

The project authority did not freeze quantitative Level-1 accuracy thresholds. The following are proposed pre-registration targets and become binding only after teacher/owner approval.

Metric	Proposed target	Why it matters
Independent evidence	At least three days with independently started normal and abnormal sessions; at least one complete day/remount held out	Prevents a single-run demonstration from being called reliable
Sensitivity to induced abnormal	≥90% on the untouched Level-1 test	Limits missed abnormal states
Specificity for normal	≥90% on the untouched Level-1 test	Limits false alarms
Stable-normal false alerts	No more than one alert during a 10-minute locked stable-normal run	Makes the live demonstration credible
Decision latency	≤2 seconds for the minimum classifier	Maintains interpretable local alerting
Decision coverage	100% for Level-1 unless an explicit sensor/invalid fault is logged	Avoids hiding errors through abstention in the semester binary task
Acquisition integrity	Zero lost analysis blocks during the locked run; saturation and alias/bandwidth limits reported	Ensures apparent accuracy is not built on missing data
Reproducibility	A second operator can execute the demo from the checklist	Shows project readiness rather than owner-only operation

10.3 Required performance reporting

- Confusion matrix and class counts, not only a single accuracy percentage.
- Sensitivity, specificity, precision where meaningful, false-alert count and missed-detection count.
- Decision latency and persistence delay.

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- ESP32 RAM/flash use, feature/FFT timing and dropped-block counters.
- Normal and abnormal plots from untouched sessions with the same axis/range.
- Failure cases, remount sensitivity and known limits of the ADXL345 frequency band.

11. Twelve-week semester roadmap

Week	Primary work	Owner(s)	Measurable output	Exit criterion	Fallback
1	Approval, scope freeze, safety and procurement	Nihad + team; Amith hardware audit	Approved scope; final order list; lab/tool inventory; risk register	No purchase until change and rig-safety direction are approved	Use available verified ESP32/motor variants without changing architecture
2	ESP32/ADXL345 acceptance and motor inspection	Sreehari firmware; Amith hardware; Sreeprada checklist	Device ID/axis test; motor/supply inspection; provisional mount plan	Sensor and motor pass acceptance	Return/replace failed module before integration
3	Stable SPI acquisition vertical slice	Sreehari + Nihad	Timestamped X/Y/Z stream; achieved ODR; loss/saturation counters	Continuous stable acquisition and USB logging	Reduce ODR/SPI clock; shorten wires; move to perfboard
4	Rigid mount and normal-state sessions	Amith rig; Sreeprada records; Nihad review	Final mount ID; normal sessions on separate starts	Feature distributions repeatable enough to define baseline	Rebuild mount/base before classifier tuning
5	Safe induced-abnormal fixture and sessions	Amith + faculty safety; team operators	Guarded eccentric fixture; repeated abnormal labels	Safe, reproducible state separation visible in raw/features	Change mass/radius/speed within safe limits
6	RMS + persistence and RGB/USB state	Sreehari firmware; Nihad integration	Working local minimum classifier and logs	End-to-end Normal/Abnormal demonstration	Simplify to one magnitude feature; improve calibration
7	Triggered FFT/bands and comparator	Sreehari data; Nihad review	FFT timing, spectra/bands; optional Mahalanobis prototype	Baseline remains stable; comparator does not break acquisition	Drop comparator if timing/data burden is excessive
8	Independent sessions and held-out protocol freeze	Nihad protocol; Sreeprada registry; Archa documentation	Locked test plan, thresholds, session split and demo draft	Minimum demonstrable system complete by end of week	Use Weeks 9-10 to stabilize baseline; no future-scope expansion
9	Edge cases and fault handling	Team	Startup/shutdown, shock, loose-mount, saturation and sensor-fault logs	Known edge cases produce honest state/fault evidence	Document limitation if reliable handling is not feasible
10	Locked Level-1 validation and demo freeze	Nihad + Sreehari; independent operator	Untouched results, metrics, resource profile and frozen release	Approved acceptance target met or failure documented without patching	No post-hoc threshold change; repeat only after formal redesign
11	Documentation and rehearsal	Archa lead; all technical review	Final dossier updates, wiring/BOM, results, demo script and limitations	Teacher-ready package; second operator can run demo	Cut nonessential content; retain core evidence
12	Contingency repair and final delivery	Full team	Stable submitted prototype and archived evidence	No new feature; deliver/rehearse	Repair from spares; present validated subset honestly

Schedule rule

The minimum demonstrable system should work by the end of Week 8. Weeks 9-12 are for validation, recovery, documentation and presentation. PIRG is prohibited from becoming a late semester feature.

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12. Team roles, actual members and accountability

Role status

The following name-to-role map is a proposed implementation assignment based on the established five-member team and earlier presentation responsibilities. The team and teacher should sign or amend it before Week 1 closes.

Member	Proposed primary role	Specific VibeGuard ownership	Review/backup obligation
Nihad P C - JEC24CC044	Technical integration lead / project manager	Architecture control, schedule, interfaces, safety coordination, validation protocol, final integration and evidence integrity	Backs up firmware and hardware; approves configuration freezes
Sreehari K - JEC24CC055	Firmware, signal-processing and data lead	ADXL345 driver, SPI acquisition, feature pipeline, RMS/persistence, FFT/comparators, USB schema and analysis scripts	Nihad reviews releases; Sreeprada executes scripted tests
Amith Krishna Das - JEC24CC016	Hardware, power and physical-rig lead	Procurement audit, motor/base/guard, rigid sensor mount, supply wiring, perfboard and safety inspections	Nihad reviews interfaces/safety; Sreehari checks signal-integrity needs
Sreeprada K S - JEC24CC056	Learner, test and inventory assistant	Inventory, configuration labels, scripted sessions, mount photos, operator checklists, run registry and repeated tests	Works from signed procedures; learns USB logging and basic firmware operation
Archa Pramod - JEC24CC022	Documentation, teacher communication and demonstration lead	Weekly review notes, teacher-facing explanations, source register, live-demo script, presentation coordination and lead communication	Technical statements reviewed by Nihad, Sreehari and Amith

12.1 Responsibility rules

- Every critical subsystem has one primary owner, one technical reviewer and one written checklist.
- No one may change hardware wiring, firmware thresholds or test labels without a dated configuration update.
- The learner/test role is substantive: repeated independent runs, inventory control and evidence integrity are project-critical.
- The documentation lead does not publish future-research details without technical and IPR review.
- Nihad acts as final integration failsafe, but work must remain distributed so the project does not depend on one member.

13. Procurement, acceptance and configuration control

13.1 Procurement sequence

1. Obtain teacher approval and freeze the semester architecture.
2. Order the ESP32, one ADXL345 module, motor/fan and power adapter first.
3. Accept or reject the sensor and motor before ordering duplicate or final fabrication items.
4. Fabricate the base, guard and rigid bracket after physical dimensions are known.
5. Buy a spare sensor or motor only when acceptance/delivery risk justifies it within the conservative envelope.

13.2 Component acceptance record

Component	Acceptance checks	Reject if
ESP32	Boots reliably, USB serial works, board identity/pinout recorded, regulator stable	Unknown board mapping, repeated disconnects, overheating or unstable supply

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13.3 Configuration ID format

Recommended format: VG-[rig]-[mount]-[sensor]-[firmware]-[protocol]-[session]. Example: VG-R01-M02-S01-FW0.6-P03-D2N04. The exact format may change, but every data file and photograph must resolve to one reproducible physical and software configuration.

14. Risk register and mitigation

Risk	Likelihood / impact	Early indicator	Mitigation	Decision consequence
Counterfeit or poor-quality ADXL345 module	Medium / High	Wrong device ID, unstable noise, limited ODR or clipping	Reputable source, formal acceptance test, preserve seller/board photos, replace before integration	Do not tune algorithms around a failed sensor
Unsafe or unstable abnormality fixture	Medium / Critical	Fastener movement, guard vibration or base walking	Retained mass, shaft guard, heavy base, low-speed/current-limited bring-up and faculty review	Stop powered testing until redesigned
Mount-dependent result	High / High	Feature shift after minor handling	Rigid documented bracket, mount ID, remount trials and photos	Report mount dependence; do not claim generality
SPI/acquisition loss	Medium / High	Timing overrun, sample gaps, corrupted frames	Short wiring, conservative clock/ODR, interrupt/FIFO use, perfboard and counters	No locked result accepted with silent loss
Sensor saturation or bandwidth mismatch	Medium / High	Clipped samples or missing discriminating frequency	Adjust range; verify signal band; limit claims; consider documented future sensor upgrade	Do not claim phenomena outside measured band
Unrepeatable normal/abnormal labels	Medium / High	Overlapping sessions or changing motor behavior	Stabilize base, speed/load and eccentric fixture before thresholding	Redesign rig, not the classifier
Data leakage / inflated accuracy	Medium / High	Random-window split gives much higher result	Split complete sessions/days/remounts; freeze thresholds before hold-out	Reject leaked metric
Late algorithm expansion	High / Medium	Core demo delayed by ML/feature work	RMS+persistence first; optional methods after Week 6	Drop comparator, preserve minimum system
Team bottleneck	Medium / High	One person owns code, rig and report knowledge	Primary/reviewer/checklist structure; weekly cross-training	Reassign before critical week
Patent expectation distorts semester	Medium / High	PIRG features enter BOM/schedule or public claims	Separate folders, budgets and claims; teacher-safe wording; IPR review	Remove future mechanism from semester build

15. Required semester deliverables and live demonstration

15.1 Deliverables

- Working guarded VibeGuard Level-1 prototype.
- Final BOM, source record, wiring/pin map and power-domain diagram.
- Motor-rig configuration and safety checklist.
- Versioned ESP32 firmware and reproducible analysis scripts.
- Raw-session register, configuration metadata and held-out validation results.
- Final report containing results, limitations, negative evidence and resource use.
- Teacher-ready presentation and live-demo procedure.
- Archived OpenBraille zeroth-review package kept separate from VibeGuard implementation evidence.

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15.2 Recommended five-minute live demonstration

1. Show the guarded rig, rigid sensor mount, ESP32 and separate power domains while power is off.
2. Connect USB, show the configuration banner and complete a short calibration/self-check.
3. Run the normal state: green LED and live USB features/logs.
4. Apply the pre-inspected induced-abnormal configuration: red LED after persistence and visible logged feature/spectral change.
5. Return to the approved normal configuration and show recovery according to the frozen state rule.
6. Show one held-out result table, resource measurements and the explicit limitations: controlled rig, Level 1 only, no patent claim.

15.3 Demo failure fallback

If the powered abnormality demonstration cannot be run safely in the classroom, use an approved guarded video of the locked setup plus a live sensor/firmware playback only if the teacher accepts that fallback. Do not improvise an unguarded rotating mass for presentation impact.

16. Post-semester PIRG research hypothesis

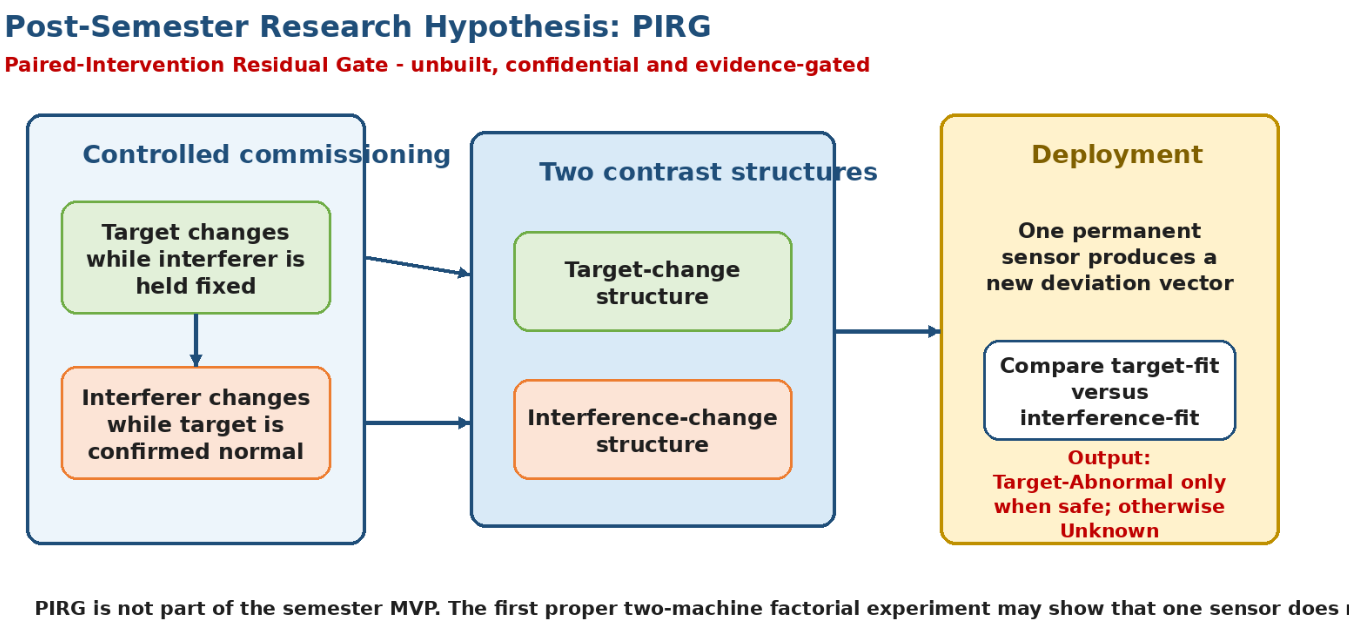

Figure 4. PIRG research concept. This is not implemented or validated in the semester system.

16.1 Technical question

Can one target-mounted vibration sensor preserve enough information to distinguish a change caused by the target machine from a change caused by an independently operating neighbouring source, when the system is commissioned using controlled paired physical interventions?

16.2 Candidate mechanism in non-claim language

1. During commissioning, change the target state while keeping the interferer at the same labelled condition. Form a target-change representation from paired differences.
2. Separately change the interferer while the target is independently confirmed normal. Form an interference-change representation.
3. At runtime, compare a new deviation against both representations using one permanent target-mounted sensor.
4. Declare Target-Abnormal only when the event is anomalous and fits the target-change structure materially better than the interference-change structure.
5. Return Unknown for ambiguous, unsupported or mount-invalid observations rather than forcing a target-fault claim.

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6. Defer quarantined baseline updates until the attribution core has independently passed; do not make every familiar feature part of one speculative bundle.

16.3 Why this is only a hypothesis

- One sensor observes a mixture; the two physical sources may be non-identifiable or nearly collinear.
- Mounting, speed, load and structural coupling may change the contrast structures.
- Known subspace comparison, open-set rejection, Mahalanobis models and single-sensor classification create high obviousness risk.
- No current experiment shows that PIRG beats strong conventional baselines.
- A model simulation cannot replace enablement, field labels or professional patent review.

17. Six-to-twelve-month future research roadmap

Month	Stage	Work	Pass evidence	Failure action
0-2	Semester foundation and research design	Freeze Level-1 platform; preserve raw data/timing/mount records; design two-motor coupled rig and temporary ground truth	Stable Level-1 data; safety-approved target/interferer scaffold	Continue semester/publication only if a reliable scaffold cannot be built
2-4	Early paired-intervention gate	Collect target normal/abnormal × interferer off/normal/abnormal across ≥3 days/remounts; compare RMS, FFT/bands, Mahalanobis and conventional classifier	Untouched day/remount shows ≥30% relative reduction in wrong attribution/false target alerts, ≤5 percentage-point miss-rate worsening and ≥80% coverage	Kill PIRG if target/interference directions are inseparable or gain is leakage/abstention
4-6	Mechanism and ablation phase	Freeze PIRG representation; remove each asserted component; test held-out interferer level, speed/load and mount validity	Complete mechanism outperforms strongest baseline and critical ablations on locked data	Remove non-contributing features; kill patent route if result is additive/generic
6-9	Field-like replication	Test more than one motor/arrangement; obtain independently labelled target/interferer states; run local ESP32 timing/resource tests	Effect survives another arrangement without permanent reference sensors	Retain only as bench publication if transfer fails
9-10	Confidential evidence freeze	Lock architecture, datasets, negative results, contributor chronology and public/private disclosure log	Reproducible evidence pack and clear claim-to-experiment map	Do not file from an outcome-only idea
10-11	Fresh claim-level search and IPR review	Refresh worldwide patent/NPL search; institutional invention disclosure; counsel-led novelty, inventive-step, eligibility and FTO screen	Named useful scope remains after closest art and design-around analysis	Trade secret/publication/kill if scope is trivial or blocked
11-12	File/no-file management gate	Select narrow supported filing, confidential continuation, publication or termination	One reasoned written decision; no grant promise	Archive and publish only after IPR/FTO review

17.1 Research baselines and ablations

Type	Required comparison
Strong conventional baselines	RMS+persistence; FFT/bands; ordinary Mahalanobis; static spectral subtraction; a competent single-sensor classifier
PIRG component ablations	Target contrast removed; interference contrast removed; Unknown forced into binary; mount-validity gate removed; operating-state conditioning removed
Evidence split	Hold out complete day/remount and at least one interference condition; never random-window leakage

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Type	Required comparison
Local deployment	ESP32 acquisition and inference with no locked-run block loss or hidden saturation
Physical truth	Temporary reference sensors or independent controls may label experiments but are not permanent product sensors

17.2 Patent-development kill conditions

- Target and interference changes occupy essentially the same observable signal direction.
- PIRG does not beat the strongest conventional comparator on held-out sessions.
- Performance works only by classifying an impractically large fraction as Unknown.
- A permanent second/reference sensor is required for operation.
- The ADXL345 misses the discriminating band and a broader sensor change destroys the connected architecture case.
- Remounting, speed/load or another motor arrangement destroys the result.
- Fresh prior art shows the claim-defining combination or useful scope becomes trivial/easily avoided.
- Institutional/professional review does not justify filing cost or disclosure risk.

18. Confidentiality, inventorship and disclosure control

18.1 Public-safe semester material

- ADXL345 + ESP32 architecture, SPI acquisition, time features, triggered FFT, RMS/persistence, ordinary Mahalanobis comparison, RGB and guarded eccentric-mass rig.
- Level-1 normal-versus-induced-abnormal results, provided limitations and configuration are disclosed.
- Budget, schedule, team roles and educational objectives.

18.2 Keep private until IPR review

- Exact PIRG paired-intervention representation, residual/fit calculation and thresholds.
- Any baseline quarantine, update, rollback or mount-validity mechanism that later survives ablation.
- Labelled interference/remount datasets, field-like results and claim-to-experiment maps.
- Unpublished negative/positive ablation results and any combination showing an unexpected technical effect.

18.3 Inventorship/contribution record

Maintain a dated record of who conceived each later mechanism, who designed each experiment, who implemented code/hardware, and when the result was first reduced to practice. Model-generated wording is not a substitute for human conception. The institutional IPR cell should determine inventorship and disclosure strategy before any filing.

18.4 Disclosure rule

Do not rely on an assumed grace period

Before publishing code, datasets, posters, videos, papers, competition entries or detailed teacher material about the future mechanism, obtain institutional IPR review. The public semester MVP and confidential future research must remain in separate folders and explanations.

19. Decision gates and escalation rules

Gate	When	Pass	Fail response
G0 - Teacher approval	Before procurement	VibeGuard Level-1 approved with scope/budget	Continue approved concept or revise request
G1 - Component acceptance	Week 2	ESP32, ADXL345, motor and supply pass checks	Replace before integration

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Gate	When	Pass	Fail response
G2 - Acquisition integrity	Week 3	Stable ODR/timestamps; visible counters; no silent loss	Reduce rate/clock, fix wiring or replace sensor
G3 - Rig repeatability	Weeks 4-5	Normal and abnormal states are safe and repeatable	Redesign mechanical fixture before algorithm work
G4 - Minimum demonstrable system	End Week 8	Local RMS+persistence, RGB and USB path works	Use Weeks 9-10 for stabilization only; cut optional features
G5 - Locked semester validation	Week 10	Teacher-approved Level-1 metrics reported honestly	Submit as controlled measurement study with documented limitations if missed
G6 - PIRG early research	Months 2-4 post-semester	Attribution improvement survives untouched session/remount	Stop patent spending; retain publication/engineering value
G7 - IPR/file-no-file	Months 10-12	Evidence, useful scope, search and professional review support action	Trade secret, publication or kill

20. Open questions and teacher decisions

ID	Question / decision	Owner	Required by
Q1	Approve VibeGuard as the semester implementation and this dossier as the controlling build plan?	Project coordinator	Before procurement
Q2	Confirm whether the implementation window is effectively 8 weeks or the full 12-week roadmap?	Project coordinator	Week 1
Q3	Approve verified ESP32-WROOM-32 DevKit or authorize ESP32-S3 based on availability?	Teacher + Nihad/Sreehari	Before order
Q4	Approve the proposed quantitative Level-1 validation targets or replace them with another pre-registered protocol?	Teacher + technical leads	Before Week 8
Q5	Which lab tools and safety reviewer are available for the motor rig?	Amith + department	Week 1
Q6	Approve the proposed name-to-role map or require changes?	Full team + teacher	Week 1
Q7	What project information may be made public before institutional IPR review?	Project owner/IPR cell	Before public disclosure
Q8	Does the department have a later field-like site or industry contact for independently labelled target/interferer research?	Coordinator/department	Post-semester planning

21. Teacher approval and conditions

Decision item	Teacher response
Semester project	Approved / Approved with conditions / Not approved
Approved board	ESP32-WROOM-32 DevKit / ESP32-S3 / Other authorized board: __________

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Approved budget ceiling	₹ _________
Approved implementation window	_________ weeks
Approved Level-1 acceptance protocol	As proposed / Modified as attached / To be submitted by _________
Safety/facility conditions	
Other conditions/comments	
Signature and date	

22. Source basis

This dossier was prepared from the controlled Project mC master package. The following files are the main evidence layers:

Evidence layer	Controlled source
Project authority	Engineering Design Review; Project mC Decision Register v1.2; approved Phase 3C Portfolio Closure Memo
Comparative evidence	Concept Evidence Matrix; Uncertainty and Test Register
VibeGuard technical authority	Final VibeGuard Architecture Report, Memory and SOP
Comparative decision context	Two final cross-concept adjudications, treated as context rather than votes
Future-research boundary	VibeGuard Pre-Build Future Patent-Case Simulation, treated as an unbuilt hypothesis
Current cost/spec basis	Project mC Teacher Documents Master Evidence, Costing and Decision Basis, price-checked 03 August 2026
Previous academic context	Final OpenBraille zeroth-review presentation and approved change request

22.1 Current official specification and price references

- Analog Devices, ADXL345 product page and data sheet: https://www.analog.com/en/products/adxl345.html
- Espressif, ESP32-WROOM-32 data sheet: https://www.espressif.com/sites/default/files/documentation/esp32-wroom-32_datasheet_en.pdf
- Espressif, ESP32-S3 product page and DevKitC-1 guide: https://www.espressif.com/en/products/socs/esp32-s3
- Indian price snapshots and procurement notes are listed in the Master Evidence, Costing and Decision Basis and must be refreshed on purchase day.

Prepared for academic project implementation. The future research section is a management and experimental roadmap, not a patent opinion, filing recommendation, freedom-to-operate clearance or promise of grant.

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